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WIDESPREAD SUBMARINE TEPHRAS AROUND JAPAN — PETROGRAPHIC AND CHEMICAL PROPERTIES

TOSHIO FURUTA[1], KANTARO FUJIOKA[1] and FUSAO ARAI[2]

[1]*Ocean Research Institute, University of Tokyo, 1-15-1, Minamidai, Nakano, Tokyo 164 (Japan)*
[2]*Department of Geology, Gunma University, Aramaki, Maebashi 371 (Japan)*

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ABSTRACT

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In deep-sea sediments around the Japanese Islands, tephra layers are frequently recognized, and some tephras can be identified by petrographic and chemical features and correlated to source volcanoes of both Japan and Korea. Most tephras distributed over more than several hundred kilometers from the source volcanoes were presumably carried away by the dominant westerly wind. Six characteristic marker tephras in deep-sea sediments can be identified and correlated according to their distribution and grain size, petrography, and chemistry. They play an important role in compiling the Late Quaternary chronology of the deep-sea sediments distributed around the Japanese Islands. Petrographic and geochemical studies reveal that the refractive index of volcanic glass shards in tephras is controlled by the amount of total transition metal oxides rather than the amount of SiO_2. We propose a new equation relating refractive index to glass shard chemistry.

INTRODUCTION

Petrographic and geochemical characteristics as well as field occurrences of tephra layers play an important role in identifying them in submarine sediments and correlating one tephra with others at remote places. Study of these characteristics also provides information on eruptions of the source volcanoes. Some widespread submarine tephra layers are geologic time markers in submarine sediments (e.g., Machida and Arai, 1976; Westgate, 1977). In the cores of deep-sea sediments, some tephra layers can be visually or petrographically recognized. Scheidegger and Kulm (1975) and Ninkovich (1979) also have shown that deep-sea tephra layers can be correlated with the source volcanoes.

Methods of investigation for tephrochronology of deep-sea sediments are essentially the same as those on land. Petrographic studies of submarine tephras have been done on grain-size and shape distribution, assemblage of constituent minerals, and refractive indices of volcanic glass shards and phenocrysts. The refractive indices of volcanic glass shards and constituent

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minerals make it possible to correlate tephra layers found at different localities with one another even when separated by a great distance (Arai et al., 1977).

The chemical composition of tephras represents the chemistry of the source volcanoes as well as magmatic processes. Chemical analyses of tephra minerals and glass shards have been done by many investigators using wet-chemical methods, electron probe microanalysis (e.g., Smith and Westgate, 1969), energy dispersive spectroscopy (EDS) (e.g., Matsuda and Kato, 1982) and atomic absorption spectrometry (e.g., Sato et al., 1983). Among constituents in tephras, the titanium content in glass shards and ferromagnetic minerals is a good indicator for correlating the tephra layers (e.g. Maenaka and Yokoyama, 1972; Shoji et al., 1975). Quenched titanomagnetites produced at the instant of tephra ejection are known to be homogeneous. Although the value is different with each sample (Momose and Kobayashi, 1972; Furuta, 1976), the Curie temperatures of these minerals contained in one layer fall into a narrow range. This is due to the homogeneity of the chemical composition of the titanomagnetite particles. Aramaki and Ui (1976) demonstrated that the Ca–Mg–Fe ratios of pyroxenes and hornblendes analyzed by EPMA proved to be a powerful tool in characterizing a set of individual pyroclastic deposits. This tool is especially useful when it is the only factor available for the stratigraphic identification and correlation of these deposits.

This paper presents petrographic data from volcanic glass shards and some phenocryst minerals found in the six submarine marker tephra layers collected from deep-sea sediments around Japan. The investigation also gives data on the refractive index and the chemical composition of volcanic glass shards from these tephra layers.

ANALYTICAL PROCEDURES

The analytical procedures for chemical compositions are as follows: Each tephra sample was purified by ultrasonic cleaning, and the clay fraction smaller than 10 μm were washed out with distilled water. Glass shards and phenocrysts were cemented in an acrylic holder with epoxy resin. Polished sections of tephras were prepared from this holder. Chemical compositions were analyzed on a real-time basis by an automated JCXA-733 (JEOL) four-channel electron probe microanalyzer, interfaced with a computer to perform a standard correction following the Bence and Albee method (Bence and Albee, 1968). The accelerating voltage of EPMA was 15 kV, and the specimen current was 12 nA on the Faraday cup. The beam diameter was 15–20 μm and major elements were counted for 10 s twice. The chemical compositions of the glass shards in each sample were averaged using five to a hundred measurements.

The experimental procedures for measurement of the refractive index of volcanic glass shards and some phenocrysts were described in Furuta and Arai (1980).

WIDESPREAD TEPHRA LAYERS

Figure 1 shows the schematic stratigraphic column of documented tephra layers occurring in both deep-sea sediments and on the islands of Japan. In this report, descriptions are given for six widespread tephra layers: the B-Tm, Ah, U-Oki, AT, Aso-4, and K-Tz in descending order. Figure 2 shows the locality map of piston cores used in this study. Petrographic and chemical features of the six tephras are summarized in Table 1. In this paper volcanic glass shards are divided into two types based on their morphology, bubble-wall and pumice types. As will be discussed in the following sections, bubble-wall glasses generally are found at greater distances from the sources than pumice glasses. Typical glass shards of both types are shown in Fig.3.

The Baegdusan-Tomakomai ash (B-Tm)

This tephra layer is distributed in an east–west zone at about 41°N (Machida et al., 1981; Machida and Arai, 1983; Fig.4a). It is intercalated in sediments at

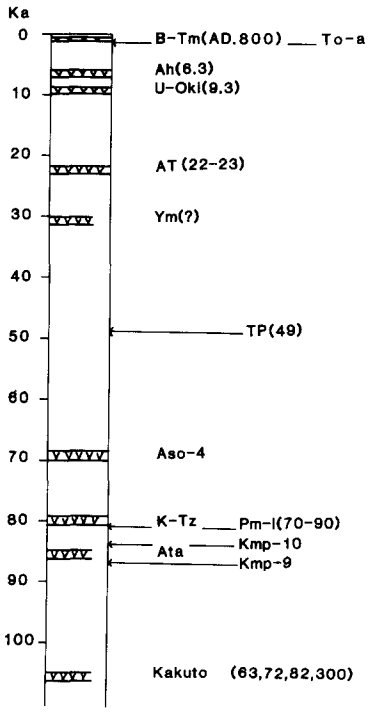


Fig.1. Schematic stratigraphic section of widespread tephra layers in deep-sea sediments of Late Quaternary age around the Japanese Islands. Important tephra layers found on the Japanese Islands are also shown. Numerals between parentheses denote ages in ka B.P.

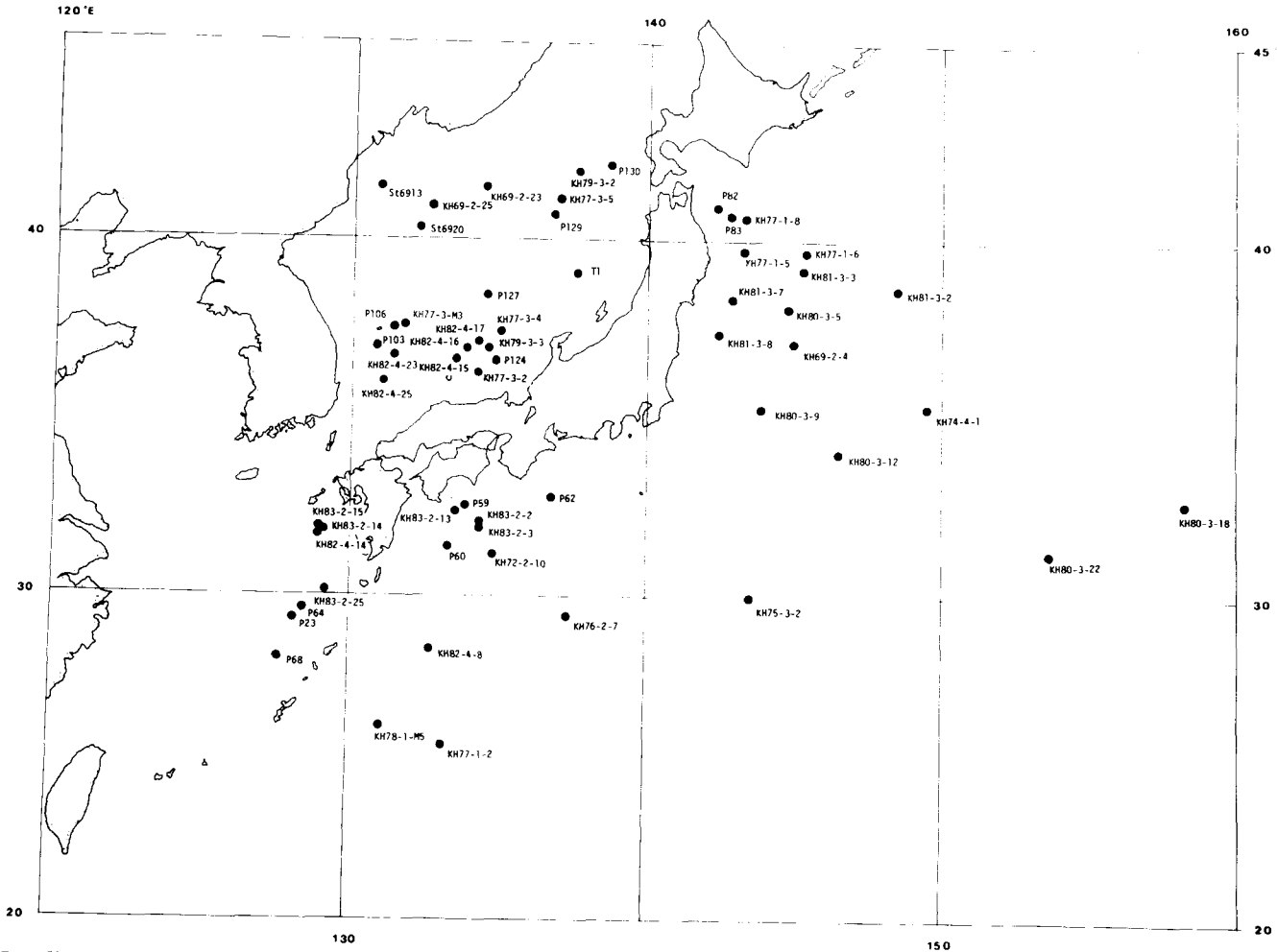


Fig.2. Locality map of piston cores used in this study. KH series were taken by R/V "Hakuho-Maru", Ocean Research Institute, University of

TABLE 1

Summary of marker-tephra layers

Tephra name	Age	Source volcano	Volcanic glass		Phenocryst minerals and refractive index
			type	refractive index	
B-Tm	800 A.D.	Baegdusan	pm≫bw	1.506–1.516	af, ho, cpx af n_1 = 1.522–1.524
Ah	6300 B.P.	Kikai	bw≫pm	1.508–1.514	pl, opx, cpx opx γ = 1.709–1.712
U-Oki	9300 B.P.	Ulreung-do	pm	1.514–1.524	af, bt, ho, cpx af n_1 = 1.521–1.524 ho n_2 = 1.730–1.741
AT	22,000 B.P.	Aira	bw > pm	1.498–1.501	pl, opx, cpx, ho, qt opx γ = 1.728–1.734
Aso-4	80,000 B.P.	Aso	bw≫pm	1.506–1.514	pl, ho, opx, cpx opx γ = 1.699–1.701 ho n_2 = 1.685–1.691
K-Tz	80,000– 90,000 B.P.	Kikai	bw > pm	1.499–1.500	opx, cpx, qt opx γ = 1.705–1.709

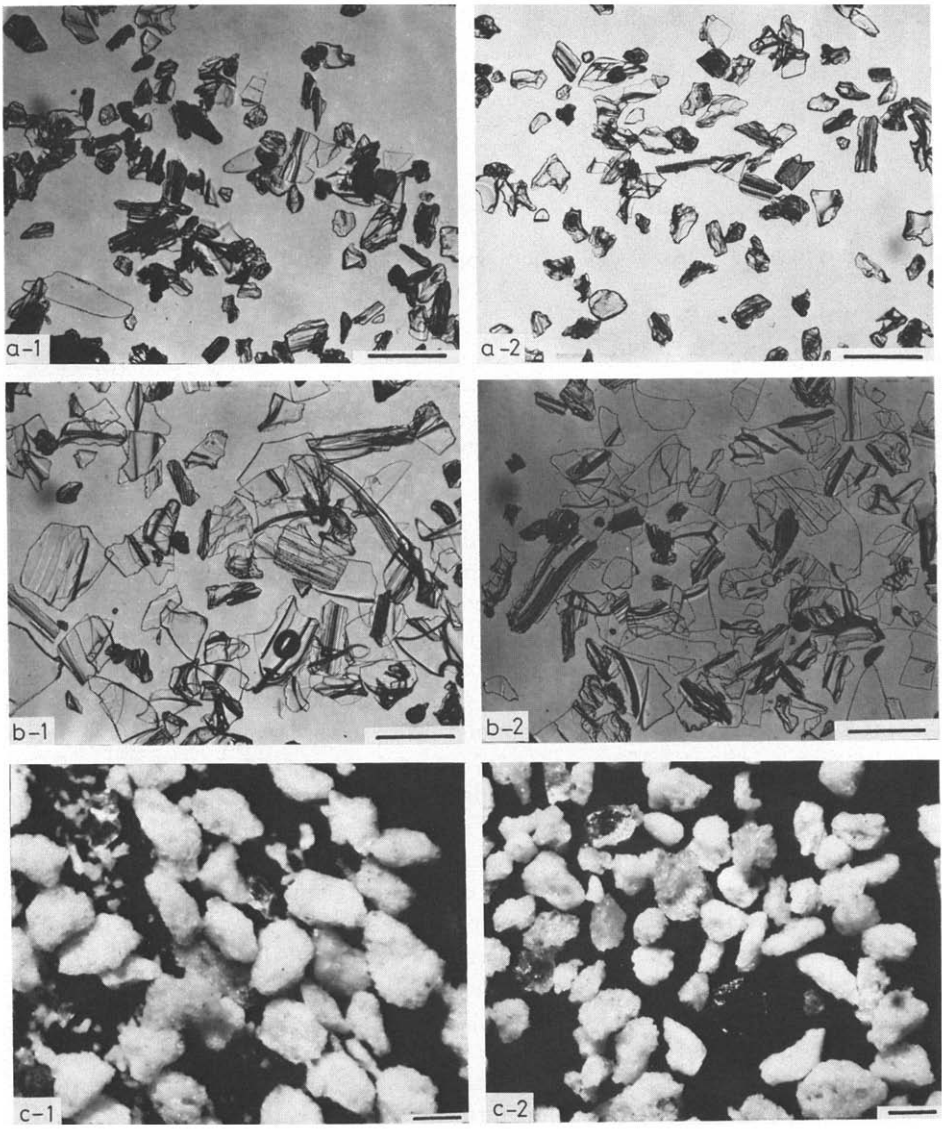
Note: pm = pumiceous shard; bw = bubble-wall shard; af = alkali feldspar; pl = plagioclase; opx = orthopyroxene; cpx = clinopyroxene; ho = hornblende; bt = biotite; qt = quartz.

or near the top of cores in the northern Japan Sea (Stn.6920, KH69-2-23, P130). On land, there are two tephra layers in humus or peat layers at historical horizons. These two tephras have several affinities in thickness and size fraction of glass shards. Judging from their distribution and chemistry, it is likely that the B-Tm layer originated from the Baegdusan in Korea. This layer consists of both pumice and bubble-wall types of glass shards. These shards are associated with a few phenocrysts consisting of alkali feldspar with cleavage flakes having n_1 about 1.523.

The refractive index of glass shards from the B-Tm layer ranges from 1.506 to 1.516. Glass shards of the B-Tm layer can be chemically classified into two groups: one with high SiO_2 content and the other characterized by low SiO_2 content. Another diagnostic chemical feature of the B-Tm glasses is their alkali content. The former group has 10–11 wt.% of total alkali, while the latter group has 8–9 wt.%. Al_2O_3 contents also show a conspicuous difference between the two groups, with the former containing about 10 wt.% and the latter about 14.5 wt.%. This suggests that there were two different volcanic products at the time of the B-Tm tephra horizon and that these originated from different fractionation stages of a magma with similar petrogenetic characteristics. The age of this layer is assumed to be 800—1000 A.D. based upon stratigraphic and archeologic data on land (Oike, 1972).

The Kikai-Akahoya ash (Ah)

This tephra is found in submarine sediment cores taken from the Japan Sea, the East China Sea, and the northern Philippine Sea as well as in land sediments all over the Japanese Islands (Fig.4b). It is considered to have been erupted from the Kikai Caldera, located south off the island of Kyushu (Machida and Arai, 1976). Glass shards in this tephra are typical bubble-walled



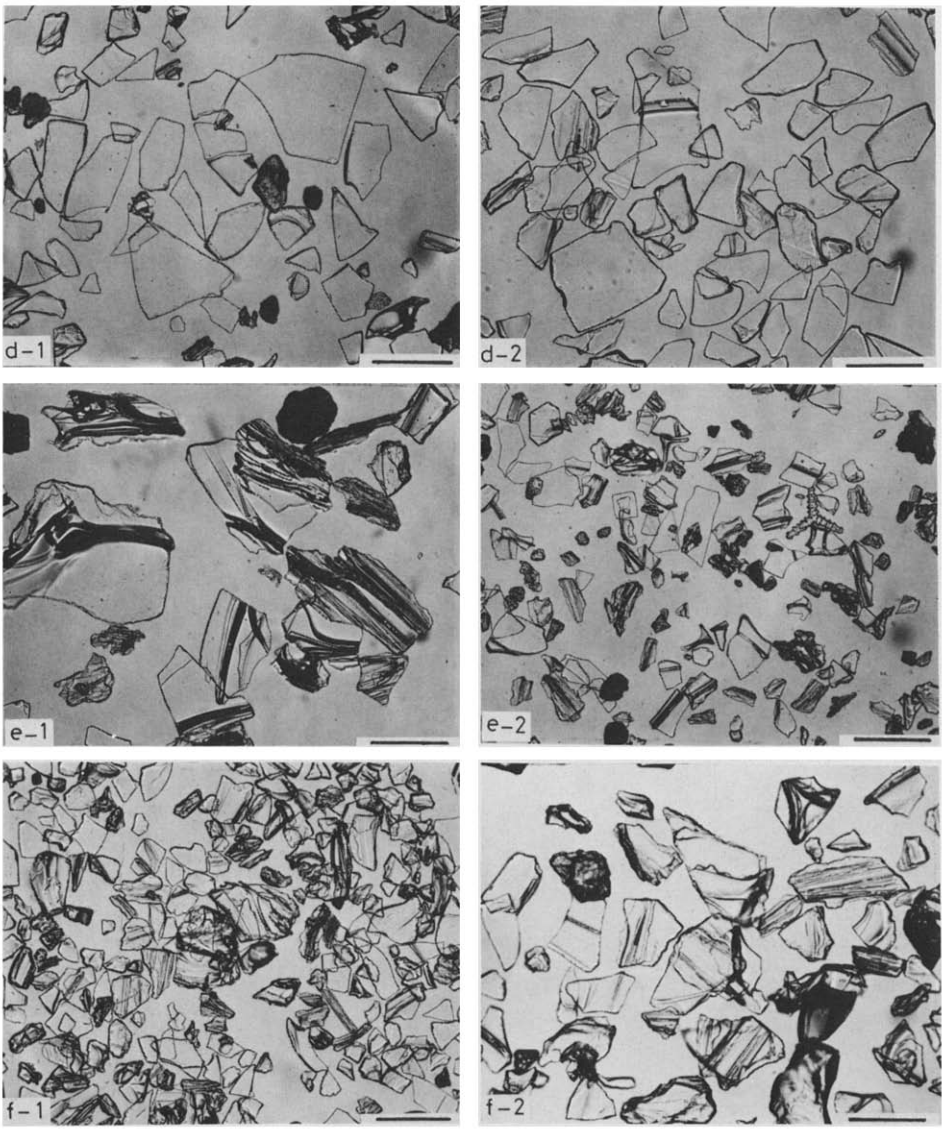
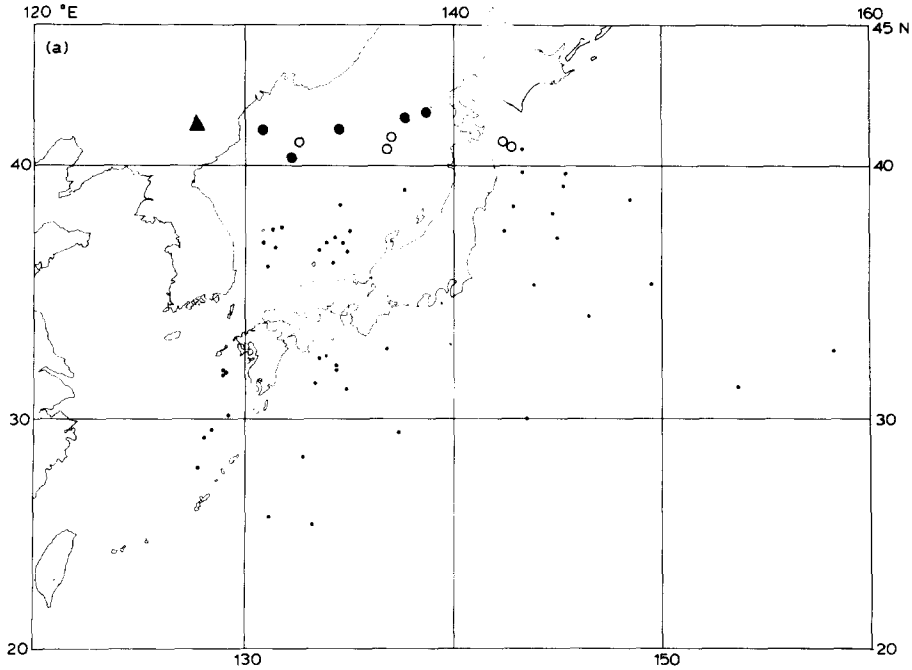


Fig.3. Photographs of volcanic glass shards. (a) B-Tm ash: a-1 from KH69-2-23, 55–60; a-2 from KH79-3-C2, 17–28. (b) Ah ash: b-1 from KH83-2-2, 300–301; b-2 from KH76-2-7, 40–42. (c) U-Oki ash: c-1 from KH82-4-15, 242–244; c-2 from KH77-3-2, 103–107. (d) AT ash: d-1 from KH82-4-14, 780–812; d-2 from KH76-2-7, 126–130. (e) Aso-4: e-1 from P106, 329–332; e-2 from KH76-2-7, 395–410. (f) K-Tz ash: f-1 from KH77-2-10, 822–829; f-2 from KH76-2-7, 425–426.5. The Ah ash and AT ash include the typical bubble-wall glass, and the U-Oki ash has the typical pumice glass. Bar on each photo is 0.2 mm.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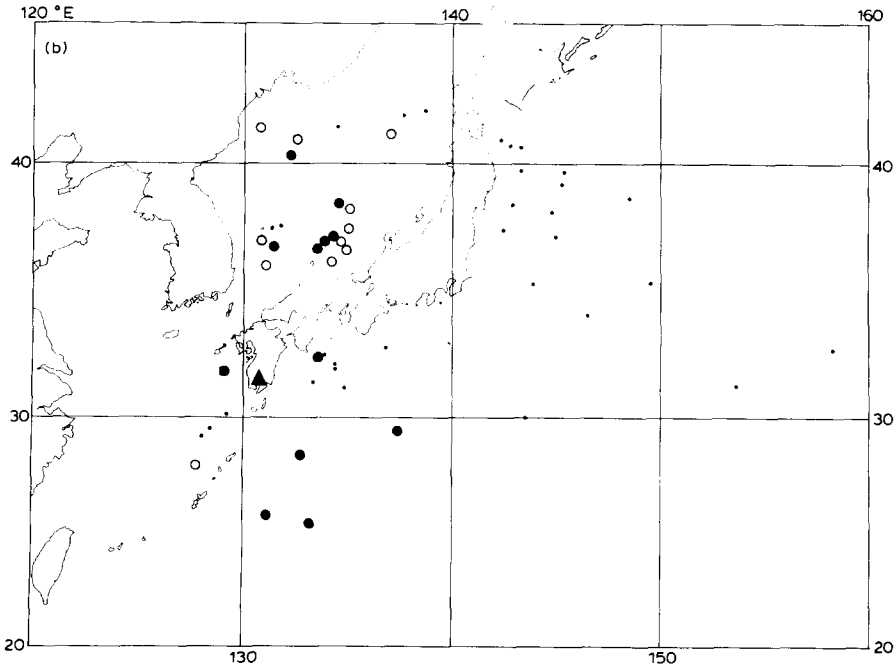
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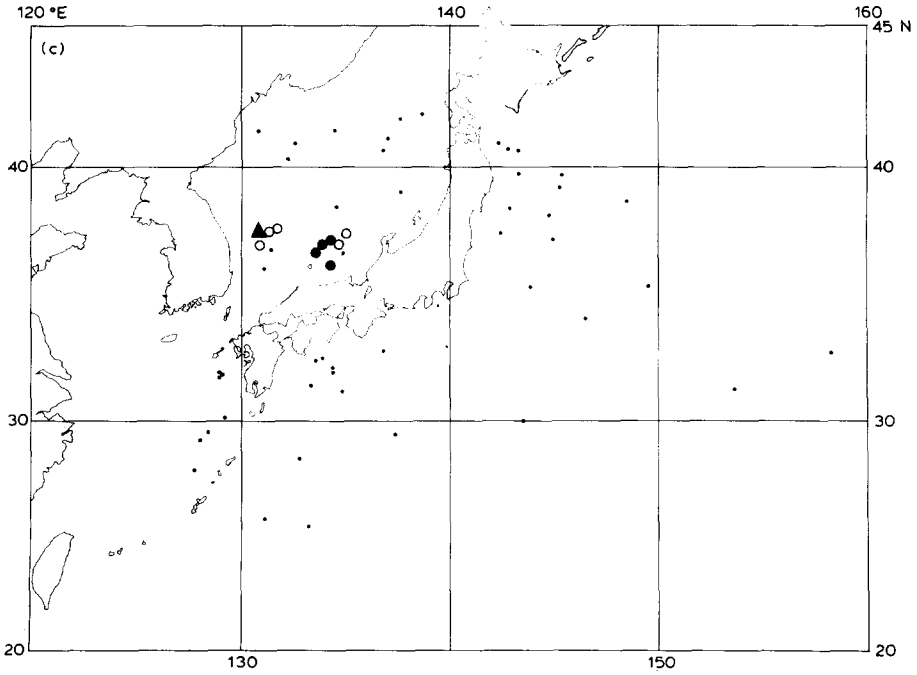


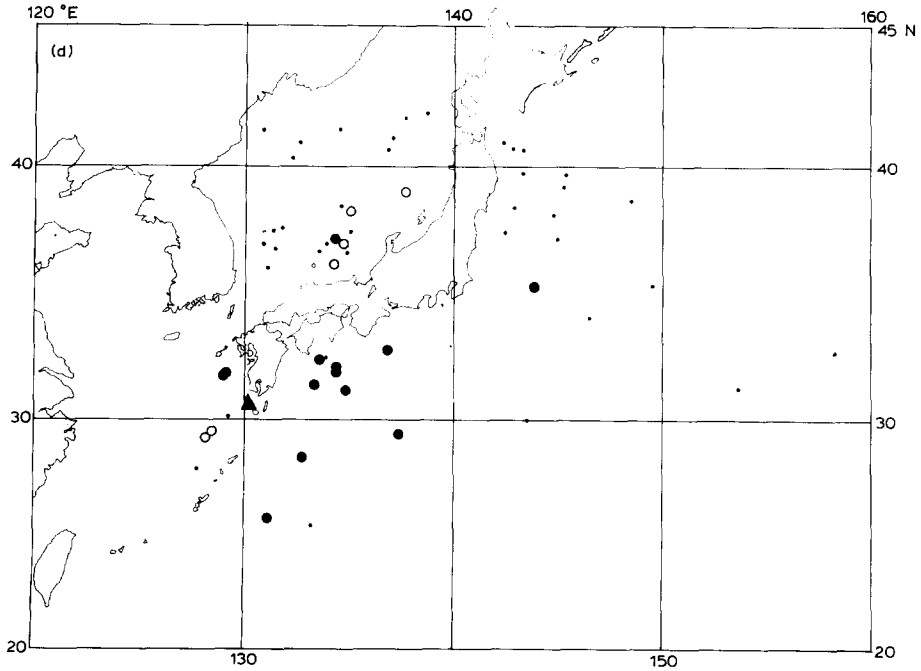
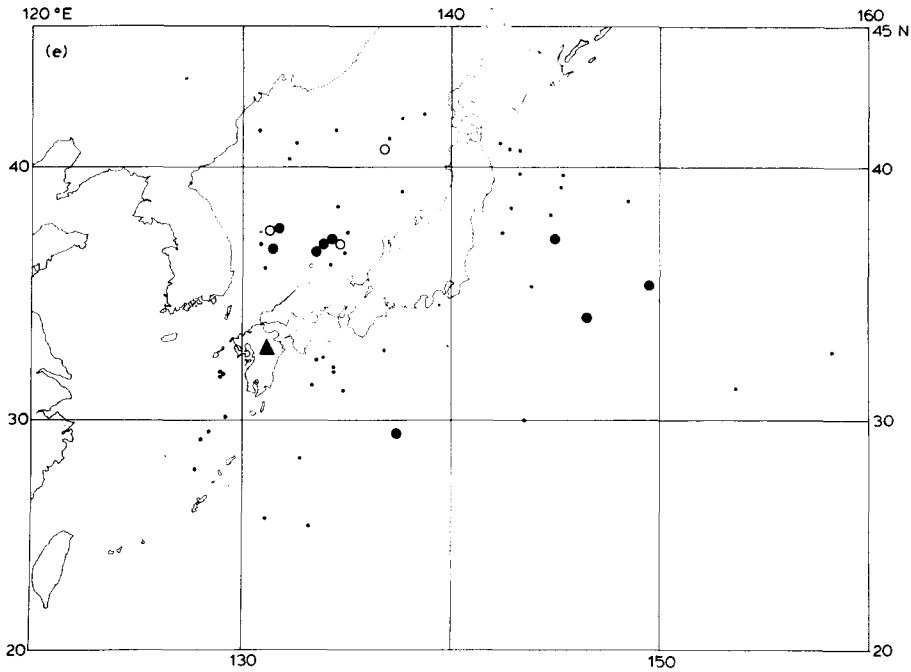


Fig.4 (a–d). *Caption on p. 135.*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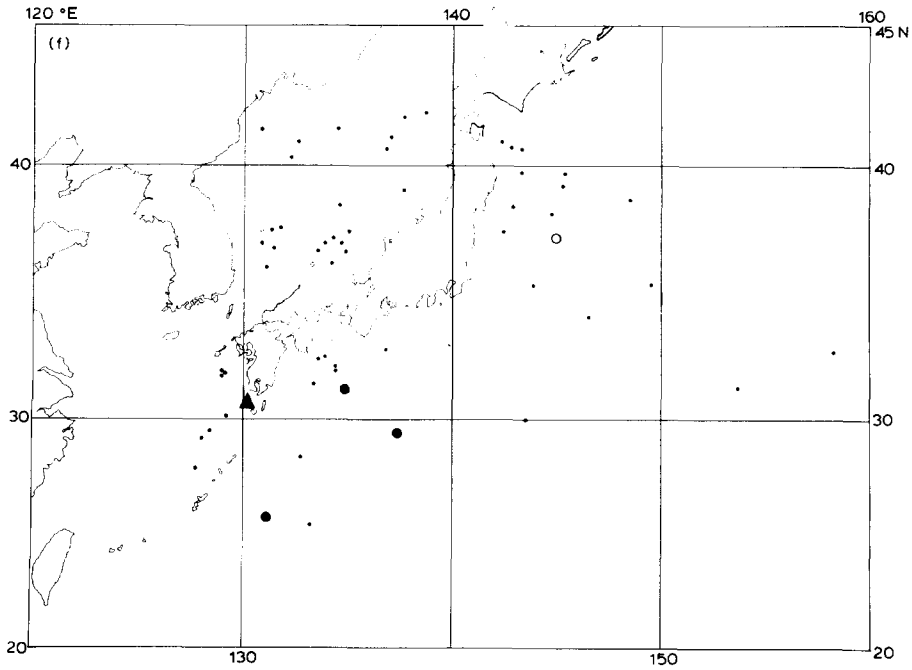
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flakes with translucent to pale brown colors. The refractive index ranges from 1.508 to 1.514. Phenocrysts consist of plagioclase, hypersthene, and augite, and γ of hypersthenes ranges from 1.709 and 1.712. Glass shards in the Ah ash are relatively low in alkali contents ($K_2O + Na_2O = 6.3$ wt.%). The contents of TiO_2, MgO and CaO are the highest of all tephra samples studied. The significance of this tephra layer is that it is intercalated in a particular horizon in the archeo-culture called the "Jomon-Culture" in Japan. The ^{14}C age is approximately 6.3 ka B.P.

The Ulreungdo-Oki ash (U-Oki)

This layer is found in submarine sediment cores from the southern part of the Japan Sea as well as from the Kinki and San'in provinces in Japan (Fig.4c). The tephra consists of pumice-type glass shards associated with alkali feldspar, cleavage flakes of which have refractive indices (n_1) which average around 1.523. Minor amounts of biotite, basaltic hornblende (approximate $n_2 = 1.740$) and clinopyroxene are also present. The refractive index of the pumice-type glass shards ranges from 1.514 to 1.524. Detailed examination of sediment cores shows that the grain size of pumice lumps decreases with increasing distances from the Ulreungdo Island. Therefore the Ulreungdo volcano is suggested to be the source. Glass shards of the U-Oki ash contain a high content of alkali ($K_2O + Na_2O = 12.4$ wt.%), and a relatively large amount of FeO* (2.7 wt.%). In a manner similar to the B-Tm ash layer, these tephras are derived from alkali rock series magmas. Such a high alkali content has not been found in any other tephra layers in the Japanese Islands.

At the Ulreungdo Volcano, three thick pumice layers, one of which may possibly be correlatable to the U-Oki ash, were recognized by Machida et al. (1984). Their chemical compositions and petrographic features are similar to those of the U-Oki ash. The chemical compositions and the refractive indices of glass shards in the three pumice layers are also similar to one another. The age of the U-Oki ash is approximately 9.3 ka B.P., based on the age of the peat deposit directly underlying these layers (Machida and Arai, 1976).

The Aira-Tn ash (AT)

This ash layer is present in submarine sediments taken from the Japan Sea, the East China Sea, the Shikoku Basin, and the northwestern Pacific as well as sediments exposed on land all over the Japanese Islands (Fig.4d). This layer has proven to be a very important key bed for identification of sea level variations in the last glacial period and for the chronology of the paleo-

Fig.4. Distribution of the six tephra layers discussed in this paper. Large solid circles denote cores analyzed by EPMA, open circles denote cores petrographically examined, and solid triangles in each figure denote the assumed source volcanoes. (a) B-Tm ash; (b) Ah ash; (c) U-Oki ash; (d) AT ash; (e) Aso-4 ash; (f) K-Tz ash.

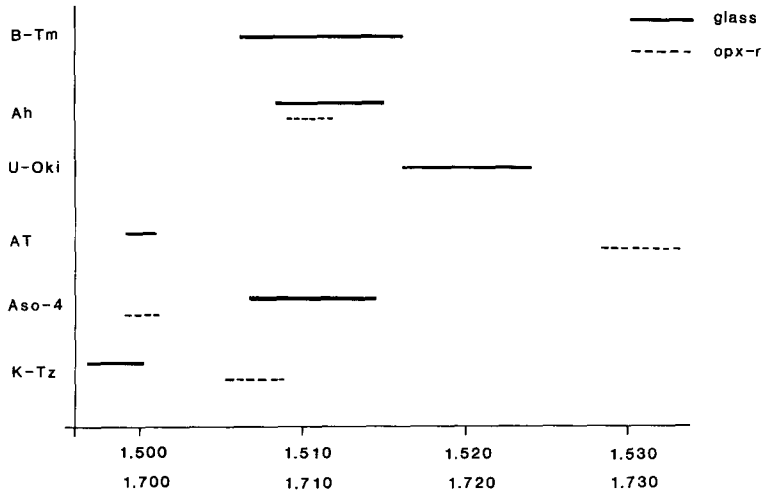

Fig.5. Refractive indices of volcanic glasses (solid line) and orthopyroxene phenocrysts (broken line) in the six tephra layers discussed in this report. *opx-γ*: refractive index of gamma of orthopyroxene.

ceramic age of Japan (Machida and Arai, 1976). The age of this layer is estimated to be 21–23 ka B.P. by ^{14}C dating of interbedded sediments (e.g., Kigoshi et al., 1972).

The petrographic nature of this tephra has been studied by several authors (e.g., Ui, 1973; Machida and Arai, 1976; Arai et al., 1977; Machida and Arai, 1983). Their results are: (1) This tephra is fine grained and well sorted; (2) the glass fraction exceeds 90% of the whole material and consists of typical plane glass and/or broken fragment of bubble-wall type; (3) the glass shards are translucent with a small deviation of refractive index ranging from 1.498 to 1.501; (4) the phenocrysts associated with the glasses are orthopyroxene, a small amount of hornblende, zircon and quartz.

The orthopyroxene phenocrysts found in this tephra are ferrohypersthene, and their refractive index ranges from 1.728 to 1.734. Such a high refractive index of hypersthene is rarely reported in Japanese tephra layers (Fig.5). The total alkali in these glasses is 6.5 wt.% and FeO* is as low as 1.25 wt.%.

The Aso-4 ash (Aso-4)

This tephra is found in submarine sediment cores from the Japan Sea and the northwestern Pacific (Fig.4e). The source volcano of this tephra is probably the Aso Volcano located in central Kyushu. In the KH80-3-12 core, the Aso-4 occurred at 143 cm from the top of the core with 2 cm thickness, and this ash layer has been recognized as the longest-distance distribution of the studied tephras (Fig.4e). Its extention is approximately 1440 km. The type of glass shards in the Aso-4 is bubble-wall, and its color is pale brown. The

TABLE 2

Chemical composition of glass shards in widespread tephras

Site[a]	SiO_2	TiO_2	Al_2O_3	FeO*	MnO	MgO	CaO	K_2O	Na_2O	Total	N[b]
B-Tm ash											
0[c]	73.61	0.22	10.44	4.17	0.07	0.02	0.24	3.71	4.86	97.50	5
1	74.07	0.21	10.20	3.97	0.06	0.01	0.20	4.11	4.37	97.20	7
2	73.62	0.22	10.44	4.17	0.06	0.01	0.25	4.14	4.61	97.53	12
3	72.50	0.23	10.27	3.98	0.02	0.09	0.27	4.34	4.53	96.25	19
3′[d]	66.83	0.37	14.49	4.27	0.13	0.11	0.97	5.20	5.32	97.73	15
4	72.90	0.22	9.86	3.91	0.02	0.07	0.23	4.11	4.46	96.00	16
4′	65.74	0.41	14.65	4.63	0.15	0.13	1.19	5.29	5.38	97.59	16
5	72.69	0.23	9.99	3.75	0.03	0.08	0.23	4.20	4.66	95.87	20
5′	67.22	0.40	13.68	4.30	0.15	0.12	0.99	4.49	5.23	96.99	9
6	71.87	0.24	10.42	3.86	0.02	0.08	0.27	3.95	4.61	95.36	25
6′	65.62	0.38	14.70	4.28	0.11	0.15	1.03	4.90	5.35	96.49	17
Ah ash											
0[c]	73.00	0.52	12.86	2.49	0.08	0.51	1.98	2.70	3.70	97.89	36
1	73.78	0.51	12.93	2.39	0.07	0.49	2.00	2.76	3.54	98.48	22
2	73.56	0.53	13.11	2.41	0.08	0.51	2.09	2.67	3.29	98.23	19
3	73.63	0.49	12.69	2.37	0.08	0.47	1.92	2.70	3.29	97.64	25
4	73.65	0.53	12.71	2.42	0.07	0.47	1.96	2.76	3.00	97.55	97
5	73.65	0.53	12.61	2.41	0.07	0.47	1.98	2.88	3.14	97.74	66
6	73.89	0.53	12.69	2.45	0.07	0.47	1.99	2.94	3.22	98.25	77
7	73.08	0.52	12.50	2.36	0.05	0.47	1.99	2.76	2.96	96.72	11
8	73.52	0.54	12.78	2.44	0.08	0.49	2.06	2.76	2.74	97.47	27
9	72.74	0.55	12.54	2.35	0.09	0.48	1.97	2.74	3.50	96.95	31
10	73.12	0.53	12.50	2.40	0.06	0.49	1.99	2.74	3.53	97.36	28
11	72.56	0.55	12.52	2.35	0.08	0.49	2.03	2.70	3.53	96.82	33
12	72.56	0.53	12.57	2.35	0.07	0.48	2.06	2.71	3.54	96.86	33
13	72.65	0.53	12.57	2.44	0.07	0.48	2.04	2.79	3.63	97.20	23
U-Oki ash											
0[c]	59.92	0.51	19.76	2.80	0.14	0.32	1.56	6.17	6.22	97.45	11
1	60.00	0.27	19.78	2.83	0.17	0.19	1.40	5.54	6.57	97.30	4
2	62.11	0.48	20.48	2.68	0.15	0.25	1.50	6.63	4.95	99.25	25
3	60.31	0.49	19.55	2.63	0.13	0.26	1.49	6.65	6.85	98.39	26
4	60.97	0.42	19.65	3.13	0.18	0.25	1.35	6.52	5.94	98.43	14
AT ash											
0[c]	75.85	0.13	11.94	1.24	0.04	0.13	1.07	3.44	3.54	97.39	46
1	75.32	0.13	11.83	1.22	0.04	0.13	1.05	3.13	3.23	96.10	22
2	75.98	0.12	11.86	1.20	n.d.	0.13	1.06	3.16	2.65	96.24	35
3	75.48	0.14	11.75	1.10	0.03	0.14	0.95	3.10	2.87	95.56	33
4	76.40	0.12	11.62	1.18	0.12	0.13	1.01	3.03	3.02	96.60	30
5	74.40	0.12	11.72	1.17	0.04	0.13	1.03	3.23	3.17	95.04	33
6	75.00	0.12	11.76	1.18	0.04	0.13	1.06	3.16	3.36	95.85	27
7	75.51	0.12	11.76	1.20	0.04	0.13	1.05	3.14	3.05	96.02	26
8	75.96	0.14	11.62	1.19	0.04	0.13	1.08	3.16	3.32	96.66	15
9	75.19	0.13	11.46	1.19	0.04	0.14	1.06	3.17	3.40	95.98	33
10	75.07	0.12	11.33	1.14	0.05	0.13	1.05	2.87	3.22	95.01	38
11	74.91	0.13	11.62	1.17	0.04	0.14	1.09	2.88	3.25	95.26	24

(*continued*)

TABLE 2 (*continued*)

Site[a]	SiO_2	TiO_2	Al_2O_3	FeO*	MnO	MgO	CaO	K_2O	Na_2O	Total	*N*[b]
Aso-4 ash											
0[c]	70.44	0.40	14.29	1.44	0.09	0.34	1.00	4.40	4.17	96.61	20
1	71.19	0.40	14.47	1.46	0.10	0.35	1.00	4.14	4.49	97.63	16
1′	70.82	0.42	15.01	1.65	0.11	0.46	1.39	3.94	4.59	97.87	7
2	70.01	0.39	14.34	1.52	0.09	0.36	1.08	4.29	4.25	96.36	20
3	70.52	0.34	14.12	1.43	0.10	0.35	1.03	4.36	4.21	96.47	30
4	69.24	0.41	14.29	1.51	0.09	0.38	1.09	4.08	4.33	95.45	23
5	69.97	0.40	14.42	1.50	0.10	0.37	1.11	4.29	4.39	96.57	30
6	71.34	0.39	14.50	1.48	0.11	0.34	0.96	4.21	4.53	97.87	21
6′	69.97	0.44	15.31	1.76	0.12	0.48	1.38	3.85	4.72	97.87	6
7	71.50	0.41	14.60	1.52	0.09	0.36	0.96	4.12	4.85	98.42	10
7′	70.25	0.42	15.20	1.76	0.11	0.49	1.36	3.88	4.91	98.42	7
8	69.59	0.42	14.29	1.57	0.09	0.40	1.29	4.28	4.44	95.39	22
8′	73.44	0.22	11.13	1.04	0.02	0.20	1.07	3.15	3.11	93.38	12
9	68.87	0.41	13.88	1.41	0.09	0.35	1.05	4.36	4.67	95.11	23
9′	67.95	0.46	14.59	1.71	0.11	0.55	1.49	4.05	4.87	95.73	12
10	68.65	0.42	14.02	1.52	0.09	0.40	1.18	4.25	4.45	96.06	83
K-Tz ash											
0[c]	73.74	0.22	11.23	1.00	0.04	0.19	0.99	3.04	2.81	93.29	34
1	72.63	0.20	10.80	1.00	0.04	0.19	0.99	2.93	2.74	91.53	38
2	73.33	0.20	10.97	1.00	0.05	0.24	1.02	2.97	2.92	92.70	9
3	73.57	0.22	11.00	1.03	0.05	0.20	1.02	2.84	3.10	93.07	21

[a]Sample sites and interval from top of core (cm):

B-Tm

1: Stn.6913, 5–15; 2: KH79-3-C2, 17–28; 3: P.130-2, 56–59, glass I; 3′: glass II; 4: KH69-2-23, 48–73, glass I; 4′: glass II; 5: Stn.6920, 32–35, glass I; 5′: glass II; 6: Stn.6920, 209–211, glass I; 6′: glass II.

Ah

1: KH82-4-8, 23–24; 2: KH82-4-14, 95–100; 3: KH82-4-17, 60–65; 4: KH83-2-2, 18–48; 5: KH83-2-2, 250–301; 6: KH83-2-3, 190–200; 7: KH83-2-13, 93–94; 8: KH83-2-14, 1168–1170; 9: KH76-2-7, 40–42; 10: KH72-2-10, 46–52; 11: P.62, 215–224; 12: P.60, 296–308; 13: P.59, 149–151.

U-Oki

1: KH77-3-2, 103–107; 2: KH82-4-16, 104–106; 3: KH82-4-17, 76–78; 4: KH82-4-15, 242–244.

AT

1: KH82-4-23, 371–374; 2: KH82-4-15, 450–464; 3: KH82-4-16, 202–203; 4: KH82-4-19, 864–869; 5: KH82-4-17, 280–290; 6: KH82-4-14, 807–812; 7: KH82-4-8, 88–90; 8: KH83-2-13, 340–341; 9: KH76-2-7, 126–130; 10: KH78-1-M5, 120–140; 11: Stn.6920, 335–338.

Aso-4

1: P.106, 329–332, glass I; 1′: glass II; 2: KH82-4-23, 847–848; 3: KH77-3-M3, 963–966; 4: KH82-4-16, 700–702; 5: KH82-4-17, 711–714; 6: P.129, 493.5–495, glass I; 6′: glass II; 7: KH80-3-12, 143–145; glass I; 7′: glass II; 8: KH76-2-7, 395–400, glass I; 8′: glass II; 9: KH69-2-4, 188–192; 10: KH74-4-1, 331–334.5.

K-Tz

1: KH72-2-10, 822–829; 2: KH76-2-7, 425–426.5; 3: KH78-1-M5, 300–315.

[b]*N*: number of analyzed grains of glass shards.

[c]Typical samples on land far from source volcanoes. *B-Tm*: Tashirotai, Aomori; *Ah*: Torihama, Fukui; *U-Oki*: Osaka Harbor; *AT*: Oguchi-bridge, Kanagawa; *Aso-4*: Shibata, Kanagawa; *K-Tz*: Aone, Kanagawa.

[d]Samples distinguished by accent mark have different chemistry of glass shards in the same tephra.

refractive index of the glass shards ranges from 1.506 to 1.514. Phenocrysts found in this tephra consist of plagioclase, orthopyroxene, with a low refractive index (γ: 1.699–1.701), brown hornblende, with high refractive index (n^2: 1.685–1.691) and clinopyroxene. The assemblage of these three indices (glass, orthopyroxene, and hornblende) makes it easy to distinguish the Aso-4 from other tephras. The chemical characteristics of the glass shards are a relatively low FeO* content (1.5 wt.%) and a relatively high alkali content ($K_2O + Na_2O = 8.8$ wt.%). By detailed examination of the chemical composition, the glass shards in this tephra can be separated into two groups according to a slight difference in Al_2O_3, FeO*, MgO and CaO contents (the site number marked by an accent in Table 2). The age of this tephra is estimated to be 70–80 ka B.P. by stratigraphic data on land (Machida and Arai, 1983) and around 80 ka B.P. by U-Th age determination (Omura, 1984).

The Kikai-Tozurahara ash (K-Tz)

This tephra is found in submarine sediment cores taken from the Philippine Sea (Fig.4f). The distribution and other informations of this tephra are insufficient to determine the absolute age. Based on related sediments of this layer, the age of this layer is close to that of the Aso-4 layer, i.e., approximately 80 ka B.P. The type of glass shards is bubble-wall whose color is translucent, and the refractive index ranges from 1.496 to 1.500. The layer consists of glass shards, a small amount of β-quartz, and hypersthene, with γ of 1.705–1.709, and clinopyroxene. Although the chemical composition of these glasses is similar to that of the AT ash layer, this layer is easily separated from the AT ash by its stratigraphic position.

RELATIONSHIP BETWEEN CHEMICAL COMPOSITION AND REFRACTIVE INDEX OF GLASS SHARDS

It is well known that the refractive index decreases as the SiO_2 content of glass shards increases. This relation is summarized as follows: with SiO_2 content ranging from 72 to 74 wt.%, the refractive index is around 1.500. With lower SiO_2 content, the refractive index increases inverse-proportionally to SiO_2 content, although the linear relation is slightly offset for SiO_2 contents between 55–65 wt.% (Williams et al., 1955).

The relationship between content of total iron oxides (FeO*) and the refractive index is quite linear (Furuta and Arai, 1980; Fig.6). We propose that the relationship is expressed by the following equation:

$$R = 0.007M + 1.495(\pm 0.005)$$

where R is the refractive index of glass shards and M is the content of total iron oxides (in wt.%).

The refractive index of glass shards is, more precisely speaking, dependent upon the total amount of transition metal oxides rather than those of SiO_2. In

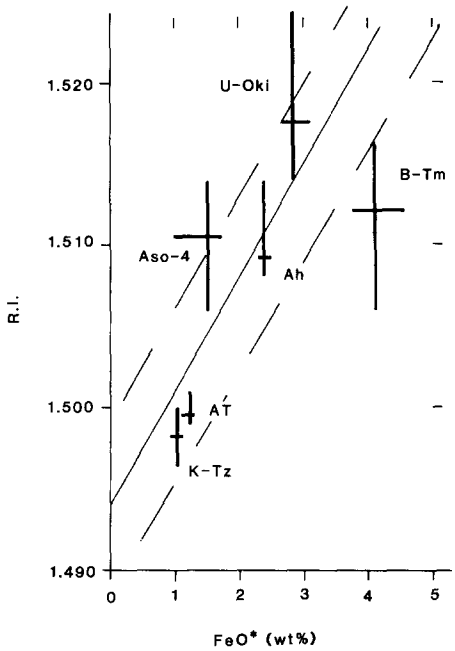

Fig.6. The relationship between refractive index of glass shards and total iron oxide (FeO*) for the six tephra layers discussed in this report. Thin solid line and broken line denote the linear trend of the refractive index as proposed in the text. Plots of high alkali glasses like as the B-Tm ash are often situated off the general linear trend (Furuta and Arai, 1980). Horizontal error bar denotes the SD of chemical analyses, and vertical bar denotes the range of the refractive index.

this report, total iron oxide is conveniently used to represent "*M*" instead of total transition metal oxides, because of less abundance of the other transition metals. Some data not on the linear trend are due to differences in the ratios of ferrous to ferric irons, because refractivity of trivalent iron ions is twice as high as that of divalent ones (Larsen and Berman, 1943).

CONCLUSIONS

The characteristics of petrographic features and chemical compositions of volcanic glass shards in the six Late Pleistocene to Holocene tephra layers found in deep-sea sediment cores and on the Japanese Islands suggest the following:

(1) Each Pleistocene to Holocene tephra has distinctive petrographic and geochemical characteristics. Several of these layers can be correlated with each other and their source volcanoes using these characteristics.

(2) Several tephras were erupted from volcanoes located on the Korean Peninsula and islands close to the Peninsula. All of them are products of extensive volcanic activity with high alkali magmas.

(3) The refractive index of glass shards is dependent on the total iron oxide (or total transition metal oxide) content rather than solely on silica content.

(4) The chemical compositions of glass shards in tephras from one series of eruptions of one volcano are very similar to each other even though the samples were collected at different localities 1000 km apart.

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